PES UNIVERSITY • DEPT. OF COMPUTER SCIENCE AND ENGINEERING

Lab 2: Epics Specification

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Problem Statement: #43 – Feature Flag & Dynamic Config Manager
Course: Software Engineering Laboratory (Lab 2)

Agile Epics Specification

This document details the functional and architectural themes (Epics) derived from Lab 1 for **Problem Statement #43: Feature Flag & Dynamic Config Manager**. Each Epic represents a significant body of deliverable functionality encompassing user stories, business value, and technical constraints.

Epic 1: Flag Lifecycle & Metadata Management (FFM-EPIC-1)

Description: Enables software engineers to create, update, and manage feature flags with globally unique identifier keys, descriptions, default boolean states, and multi-environment overrides (Dev, Staging, Production). This prevents cross-environment configuration drift and provides a single source of truth for feature states.

Business Value: Decouples code deployment from feature release, enabling safe canary releases and dark launches without requiring code rollbacks.

Associated Requirements: FR-002, FR-004 | **Traceable Use Cases:** UC-01 (Create Feature Flag), UC-04 (Override Environment Config)

Epic 2: Targeting Rules & Gradual Rollouts (FFM-EPIC-2)

Description: Provides advanced rule evaluation mechanisms allowing engineers to segment users based on demographic attributes (e.g., beta cohort membership) and configure deterministic percentage-based rollouts (0–100%) calculated via a mathematical hash of the User ID.

Business Value: Limits blast radius during feature rollouts and empowers product teams to run safe A/B tests and early access programs.

Associated Requirements: FR-001, FR-003 | **Traceable Use Cases:** UC-02 (Configure Targeting Rules), UC-03 (Evaluate Feature Flag)

Epic 3: Governance, Approval Workflows & Audit Logging (FFM-EPIC-3)

Description: Establishes release governance by requiring a 4-eyes approval workflow where Release Managers inspect configuration diffs before approving changes to production. Maintains an append-only, tamper-proof audit log capturing author, approver, timestamp, and previous state with one-click rollback.

Business Value: Protects production stability against unauthorized or erroneous configuration shifts and ensures compliance with enterprise change control standards.

Associated Requirements: NFR-002 | **Traceable Use Cases:** UC-06 (Approve & Publish Flag Change), UC-07 (View Audit Log / Rollback)

Epic 4: High-Performance Evaluation & Real-Time SDK Sync (FFM-EPIC-4)

Description: Implements a sub-15ms p95 evaluation engine backed by in-memory indexing, complemented by a persistent WebSocket streaming architecture that pushes updated flag states to connected client SDKs in real time without requiring application restarts.

Business Value: Guarantees zero latency penalty on critical application paths and enables instantaneous global kill-switch capability during production incidents.

Associated Requirements: FR-005, NFR-001 | **Traceable Use Cases:** UC-05 (Sync Flags to Client SDK), UC-03 (Evaluate Feature Flag)

Epics Summary Table

Epic ID	Summary	Story Points	Scope	Sprint Allocation
FFM-EPIC-1	Flag Lifecycle & Metadata	8 PTS	Flag CRUD, Environment Overrides	Sprint 1 & Sprint 2
FFM-EPIC-2	Targeting & Percentage Rollout	10 PTS	Deterministic Hash, Attribute Rules	Sprint 1
FFM-EPIC-3	Governance & Audit Logging	13 PTS	Approval Gate, Diff Preview, Rollback	Sprint 2
FFM-EPIC-4	Evaluation Engine & SDK Sync	16 PTS	Sub-15ms p95, WebSocket Stream	Sprint 1 & Sprint 2